ZKBlockchain

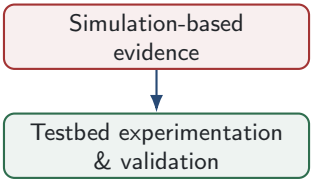
A Privacy-Preserving Blockchain Framework for Healthcare Data Sharing Using Post-Quantum Cryptography and Zero-Knowledge Proofs

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Fall 2026 • FYP-I Proposal Defense

Problem Statement

STREAM: RESEARCH-DERIVED



Existing simulation-based blockchain architectures for Internet hospital systems demonstrate promising performance, but their security, privacy, scalability and reliability claims remain **insufficiently validated** under real cryptographic implementations, distributed network conditions and realistic physiological data.

Stakeholders & Current Data Workflow

Primary Stakeholders

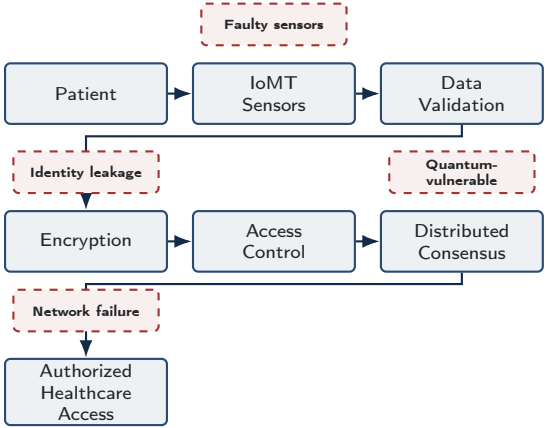
- Doctors / Healthcare professionals – authorized, timely access
- Patients – privacy & data integrity
- Hospital / Healthcare Institutions

Secondary Stakeholders

- Blockchain Administrator

SDG 3 – Good Health & Well-Being

SDG 9 – Industry, Innovation & Infrastructure



Existing Solution & Evidence-Based Gap

WHAT THE PAPER SHOWS → **WHAT IT DOES NOT YET ESTABLISH** → **WHAT OUR FYP WILL TEST**

Capability	Existing Paper	Gap	Proposed FYP
Post-Quantum Cryptography	Kyber-inspired	No standard PQC	Standardized PQC
Access Privacy	Hash commitment	No true ZK proof	Zero Knowledge Proof
Consensus	Simulated Raft	No real deployment	Multi-node HotStuff
Network	Constant delay	Unrealistic WAN	Delay, loss & failures
Sensor Data	Synthetic	Limited realism	Framingham Dataset
Security	Theoretical claims	Limited validation	Empirical threat testing

*Note: gaps are framed as **validation and implementation limitations**, not as invalidating the original paper's contribution.*

Research & Technical Basis

1. IoMT Data Reliability

- Noise, missing and spoofed readings
- Validate using realistic physiological data

2. Post-Quantum Cryptography

- RSA vulnerable to quantum attacks
- Standardized PQC over ad-hoc simulation
- Candidate: *ML-KEM*

3. Privacy-Preserving Access

- Hash commitment $\neq$ true ZK proof
- Minimize identity/role exposure
- Candidate: *Zero-Knowledge Proofs*

4. Distributed Consensus

- Raft (CFT) vs. BFT alternatives
- Require Byzantine fault tolerance
- Candidate: *HotStuff*

Decision Table

Technical Question	Alternatives	Selected	Reason
PQC primitive?	ML-KEM / ML-DSA	*ML-KEM*	NIST-standardized, quantum-resistant
Privacy mechanism?	RBAC / Hash / ZK	*ZK Proof*	Verification without identity exposure
Consensus?	Raft / PBFT / HotStuff	*HotStuff*	Byzantine resilience at feasible scale

Why This Is a Complex Computing Problem

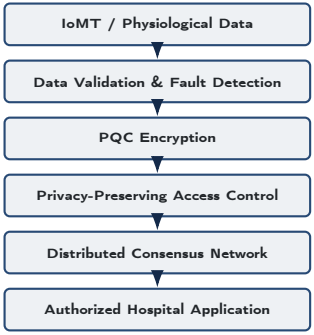
This is not routine CRUD / system-integration work. It requires balancing conflicting engineering trade-offs:

- **[T1] Security ↔ Performance** — stronger crypto/privacy increases computation & latency
- **[T2] Privacy ↔ Verifiability** — prove authorization without exposing identity/policy
- **[T3] Scalability ↔ Fault Tolerance** — more nodes change communication overhead & failure behavior
- **[T4] Reliability ↔ Complexity** — faulty sensor data cannot simply be trusted and stored

Seoul Accord Graduate Attributes Addressed

Complex problem analysis • Design under uncertainty • Trade-off evaluation • Integration of multiple technical domains (cryptography, distributed systems, networking, access control) • Experimentation and empirical validation

Proposed Contribution: System Architecture



In Scope

- Real PQC implementation
- Privacy-preserving authorization
- Distributed consensus framework
- Realistic network simulation/emulation
- Physiological/IoMT dataset validation
- Measurable benchmarking
- Testbed Implementation

Out of Scope

- Clinical diagnosis
- Production-scale infrastructure
- Regulatory certification claims
- Clinical efficacy claims

Initial Technical Spike — To Be Completed Before Final Proposal Approval

Risk Tested

Can real PQC key exchange operate together with multi-node distributed communication under non-ideal network conditions? This is the highest-risk assumption underlying the whole architecture.

Planned Experiment

- Implement PQC key encapsulation/decapsulation (candidate: ML-KEM/Kyber)
- Deploy ≥ 3 nodes on separate processes/containers
- Emulate latency/packet loss between nodes (e.g. `tc netem`)
- Measure round-trip crypto + propagation latency

Evidence (Planned)

- Console log of successful encapsulation/decapsulation
- Node-to-node message propagation trace
- Latency-vs-packet-loss benchmark chart

What We Will Learn / Design Decision

Whether the candidate PQC + consensus combination is computationally feasible at the node counts and network conditions targeted for FYP-1, or whether the scope must be adjusted (e.g. smaller cluster, alternate primitive).

No fabricated results are shown — this spike will be executed during early FYP-I

Evaluation Plan

Research Question	Baseline	Experiment	Metric	Success Criterion
Does real PQC give acceptable performance?	RSA-2048	Encryption workload	Latency, throughput, ciphertext size	[TO BE BASELINED DURING FYP-1]
Does the privacy mechanism reduce information exposure?	RBAC / hash commitment	Access requests	Proof time, leakage indicators, correctness	No unauthorized disclosure
Does consensus remain usable under realistic networks?	Baseline consensus	Delay / loss / partition scenarios	TPS, latency, finality, recovery	[TO BE BASELINED DURING FYP-1]
Does sensor correction generalize beyond synthetic data?	Raw / no correction	Real / open physiological data	Accuracy, MAE, fault detection	[TO BE BASELINED DURING FYP-1]

Non-Functional Criteria

- Recovery time after node failure
- Max. acceptable consensus latency
- CPU / memory overhead
- Network bandwidth / message overhead
- Cryptographic operation latency

Critical Failure Cases Tested

- Node crash • Network delay • Packet loss
- Network partition • Invalid authorization
- Replay attempt • Faulty/spoofed sensor reading
- Excessive sensor faults

Scope, Risks & Dependencies

Committed Scope

- **Real PQC key exchange implementation**
- **Privacy-preserving authorization prototype**
- **Multi-node distributed consensus deployment**
- **Realistic network emulation experiments**
- **IoMT fault detection & real dataset validation**

Optional Extensions

- **Full WAN (multi-region) deployment**
- **Second independent physiological dataset**
- **Formal / mechanized proof of the ZK protocol**

Key Risks → Mitigation / Fallback

- ZK integration too complex → **well-defined privacy-preserving proof prototype, benchmarked against original hash-commitment**
- Real physiological dataset unsuitable → **use a second open dataset, same evaluation protocol**
- Full WAN deployment infeasible → **controlled network emulation across multiple distributed nodes**
- PQC / distributed debugging complexity → **start with reduced node count, scale incrementally**

Dependencies: compute resources • open datasets • modern PQC/ZK libraries • containerized/network-emulation tooling

Team Responsibilities & FYP-I / FYP-II Plan

Hannan Abid	Daniyal Qureshi	Zia Ul Baqa
Post-quantum cryptography implementation & secure data storage	**Privacy-preserving / zero-knowledge access control & security evaluation**	**Distributed consensus, network realism, IoMT data pipeline & fault tolerance**

Shared: system integration • testing • benchmarking • documentation • security review • final demo

FYP-I

Literature & gap validation → requirements → architecture → technical spike / POC → baseline implementation → dataset preparation → initial experiments

FYP-II

Full integration → realistic network experiments → security/privacy evaluation → sensor validation → performance benchmarking → final refinement → demonstration & thesis

FYP-I proves feasibility; FYP-II proves the contribution.

Ethical Constraints & Legal Considerations

ETHICAL CONSTRAINTS

1. **Sensitive data, properly bounded**
   Framingham Heart Study data is used only in its de-identified, publicly released research form, under its own usage license – no live-patient data is collected.

2. **No clinical authority claimed**
   The system is never presented or used as a diagnostic or clinical decision-support tool; all outputs remain within an academic-prototype scope.

3. **Responsible use of security research**
   PQC/ZK components are used strictly for defensive privacy protection; any vulnerability found is documented for evaluation, never exploited or disclosed irresponsibly.

LEGAL & REGULATORY RISK

4. **Data-protection compliance**
   Healthcare data sharing engages frameworks such as Pakistan's Personal Data Protection Bill, alongside GDPR/HIPAA norms where comparable data is referenced.

5. **Immutability vs. erasure rights**
   Append-only ledgers can conflict with a data subject's "right to be forgotten" – mitigated by storing only encrypted references/proofs on-chain, never raw records.

6. **Cybercrime law & licensing**
   Design and testing stay within Pakistan's PECA 2016 (unauthorized access/data interference), and all third-party PQC/ZK libraries and datasets retain their license terms.

How This Is Addressed in the FYP

Evaluated strictly as an academic prototype on de-identified/public data, with supervisor and departmental ethical review sought before any real-patient extension – no production or clinical deployment is claimed.